

GREEN BAY STRAUBEL, WI, USA

WMO: 726450

Lat: 44.479N

Lon: 88.137W

Elev: 687

StdP: 14.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14898

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-3.2	-17.5	2.2	-7.2	-12.4	2.9	-2.6	25.7	19.3	23.2	19.3	8.3	290	0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	87.9	73.7	85.0	71.9	82.5	70.3	76.3	84.6	74.4	82.0	72.5	79.7	10.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.6	128.4	81.6	71.7	120.2	79.6	69.9	112.8	77.4	40.2	84.9	38.3	82.0	36.6	79.6	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.0	20.3	18.3	-15.2	93.1	5.5	3.7	-19.2	95.7	-22.4	97.9	-25.5	99.9	-29.5	102.6		
			-15.7	80.0	5.4	2.2	-19.6	81.6	-22.7	82.9	-25.8	84.1	-29.7	85.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	17.9	21.0	32.0	44.4	56.3	66.4	70.4	68.3	61.2	49.0	36.3	24.3
	DBStd	20.43	11.93	11.44	11.18	8.70	8.25	7.03	5.70	5.31	7.73	8.47	9.22	10.97
	HDD50	3959	994	812	567	211	29	0	0	0	7	125	418	796
	HDD65	7519	1459	1232	1022	620	294	66	15	27	162	501	860	1261
	CDD50	2415	0	0	10	42	224	492	633	568	344	93	8	1
	CDD65	500	0	0	1	1	24	107	184	130	49	4	0	0
	CDH74	4388	0	0	5	24	275	975	1662	991	413	43	0	0
	CDH80	1170	0	0	0	2	65	288	508	213	88	6	0	0
Wind	WSAvg	8.6	9.5	9.5	9.5	10.5	9.3	7.7	6.9	6.3	7.2	8.6	9.1	9.0
Precipitation	PrecAvg	29.6	1.3	1.1	1.9	2.7	3.1	3.7	3.3	3.4	3.2	2.4	2.0	1.6
	PrecMax	39.3	3.7	2.8	4.7	5.9	8.3	10.3	9.5	9.1	7.8	5.2	5.3	5.7
	PrecMin	17.9	0.1	0.0	0.2	0.5	0.1	0.3	0.8	0.6	0.3	0.4	0.1	0.1
	PrecStd	4.4	0.7	0.6	1.1	1.1	1.7	2.0	1.8	1.8	1.8	1.3	1.2	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.2	51.2	70.1	77.9	86.1	91.5	93.0	88.6	87.1	79.6	65.9	53.7
		MCWB	41.1	46.4	59.3	62.5	69.0	74.7	78.0	74.7	72.0	66.6	56.7	51.7
	2%	DB	39.7	43.3	59.8	70.6	81.2	86.6	88.3	85.0	82.5	73.0	58.2	45.4
		MCWB	36.5	39.2	53.0	57.9	66.1	72.4	74.7	73.1	70.0	63.4	51.9	42.0
	5%	DB	36.6	39.1	52.6	64.3	76.4	83.2	85.2	82.2	78.9	68.1	54.2	41.2
		MCWB	34.6	35.9	46.4	53.3	63.7	70.1	72.5	71.0	67.9	60.1	48.6	38.8
	10%	DB	34.2	36.2	46.8	59.5	71.7	79.8	82.5	79.6	75.3	63.1	49.8	37.8
		MCWB	32.7	33.5	41.6	49.8	60.7	68.2	70.8	69.1	65.9	57.2	45.5	35.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.8	46.8	61.0	64.3	71.9	77.9	80.3	77.6	75.0	70.1	58.1	52.0
		MCDB	44.9	51.0	69.6	73.7	81.6	87.5	90.4	84.9	83.1	76.5	63.7	53.7
	2%	WB	37.1	39.4	53.7	59.4	68.6	74.8	77.0	75.2	72.2	65.3	53.4	43.0
		MCDB	38.9	42.1	58.7	68.4	77.6	84.1	85.6	82.3	79.3	70.9	57.5	44.7
	5%	WB	34.8	36.3	46.8	54.6	65.5	72.3	74.8	73.1	69.9	61.4	49.4	38.9
		MCDB	36.4	38.8	51.6	63.1	74.7	80.5	82.3	79.4	76.2	67.2	53.4	41.0
	10%	WB	32.7	33.7	41.8	50.5	62.2	69.8	72.8	71.2	67.6	57.5	45.9	35.4
		MCDB	34.3	35.9	46.8	58.1	70.5	77.3	80.0	77.3	73.4	62.7	49.7	37.3
Mean Daily Temperature Range	5% DB	MDBR	14.0	15.2	16.1	18.6	20.2	19.4	20.3	19.7	20.9	17.8	14.2	12.9
		MCDBR	13.9	15.0	22.5	28.2	26.1	22.8	22.5	21.9	24.4	24.2	20.7	15.2
		MCWBR	12.3	12.1	15.8	17.5	14.1	12.0	11.5	11.7	13.3	15.3	15.2	12.8
	5% WB	MCDBR	13.2	14.0	22.0	26.3	23.4	20.5	20.2	18.7	20.8	21.6	19.0	14.1
		MCWBR	12.1	11.8	16.4	17.5	13.7	12.2	11.7	11.2	12.7	15.7	15.4	12.5
Clear-Sky Solar Irradiance	taub	0.269	0.320	0.343	0.376	0.406	0.412	0.398	0.398	0.371	0.330	0.300	0.275	
	taud	2.410	2.241	2.326	2.296	2.258	2.280	2.325	2.318	2.382	2.495	2.506	2.449	
	Ebn at Noon	274	272	282	281	274	272	275	271	269	269	260	260	
	Edh at Noon	23	33	35	39	42	41	39	38	33	26	21	20	
All-Sky Solar Radiation	RadAvg	499	830	1198	1435	1736	1887	1938	1652	1307	823	520	374	
	RadStd	56	78	119	144	138	135	87	109	101	82	49	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page